

Amirhossein Yousefiramandi

Applied Scientist · LLMs / NLP / Generative AI

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SUMMARY

Applied scientist with 6+ years productionizing NLP / LLM research at scale — pretraining domain-specific language models, designing taxonomy-guided classifiers, and shipping research-grade systems on AWS. Currently **Senior Machine Learning Engineer at IQVIA** (Barcelona), owning the production ranking models, data pipeline, evaluation framework, and monitoring behind **clinical-trial site tiering**; previously nearly three years at Clarivate on patent-domain NLP and generative AI. **First-author on four arXiv preprints** (ModernBERT for patents; causal-LLM classification; synthetic-patent-data study; patent-embeddings benchmark) and **sole-author on two 2026 preprints** (Forward-Forward layer free-riding; DRL for deadline-constrained coded caching); three patents (sole inventor on one, first inventor on one, co-inventor on one); two published, one pending USPTO applications. Comfortable owning the loop from hypothesis → controlled experiment → ablation → deployed service.

EXPERIENCE

IQVIA

Senior Machine Learning Engineer

Barcelona, Spain

Jun. 2026 – Present

- Own the production ML system behind **clinical-trial site tiering** end to end — a family of ranking models (site **participation**, **performance**, and **delivery**) plus the data pipeline that feeds them — used by clinical-trial teams to select investigator sites for studies.
- Build and operate the automated **model-evaluation framework**: it discovers candidate models in **MLflow**, runs bulk predictions, and ranks each candidate against the current production model on versioned datasets and ranking metrics, making promote-or-hold an evidence-backed decision rather than a judgement call.
- Guard the measurement itself: drift and **feature-space-collapse detection** against production, plausibility ceilings on candidate predictions, and de-duplication of identical retrains, so no model is promoted on a broken comparison.
- Make model quality a live surface: **Datadog dashboards defined as code** — versioned in the repo and provisioned through **Terraform**, never hand-edited — fed by metric-push steps inside the evaluation pipeline, with alerting on regression, drift, and evaluation runs that report nothing at all.
- Moved nightly data ingestion onto orchestrated **Airflow** DAGs with a real dependency graph, retries, data-quality gates, and end-to-end failure alerting, replacing fragile cron-offset scheduling and **cutting the nightly window from ~9 h to ~5 h**.
- Operate a **protocol-similarity retrieval service** that surfaces the most similar prior trials; drive documentation, dependency and vulnerability hygiene, and mentoring across the stack.
- Data-centric model work: improving the data a **two-tower ranking network** consumes rather than swapping architectures, with every change validated through the evaluation framework.

Clarivate

Senior Data Scientist (Applied Science)

Barcelona, Spain

Aug. 2023 – Jun. 2026

- Designed and shipped the core ML classifier behind **Derwent AI Classifier** (public product) — **\$10M+ in incremental revenue** and a **+30% increase in customer renewal contracts**.
- Pretrained three domain-specific masked language models from scratch on a curated 60M+ patent corpus (ModernBERT architecture, custom tokenizer, FlashAttention/RoPE/GLU, CUDA optimizations); the resulting **proprietary patent-domain LLM** outperforms general ModernBERT on 3/4 internal benchmarks at >3× the inference throughput of PatentBERT (arXiv:2509.14926).
- Co-authored a study on instruction-tuned vs. embedding-based fine-tuning of decoder-only LLMs for patent text classification; the embedding-based variant matched domain-specific BERTs while training on a single GPU under 4-bit quantization + LoRA (arXiv:2512.12677).
- Designed and productionized a multi-agent patent novelty / prior-art research system on AWS AgentCore (DeepAgents harness; DSPy / Strands / Google ADK custom agents): specialized agents call semantic patent search, keyword search, and NPL (scientific literature) tools, with input/output guardrails against prompt injection and data leakage — replacing a manual examiner workflow. The capability **secured engagement from several major enterprise customers** and **materially lifted our new-contract acquisition rate**.
- Led the data-science side of the novelty / prior-art research product, **supervising a junior data scientist** and owning end-to-end deployment via **Jenkins CI/CD**.

- Built a comprehensive **evaluation framework** for the novelty-search system on **DeepEval** — developed custom retrieval / relevance metrics complementing LLM-as-judge, plus agent-trace replay and a per-release regression suite — making improvements measurable and customer-reportable.
- Built the ML core of **Derwent AI Classifier**: mapped each customer's patents to their **customizable taxonomy**. For customers with labeled data, it trained **supervised hierarchical classifiers on demand** — one model per parent node of the taxonomy — with bespoke load-control to scale the resulting per-customer model fleet. For customers **without labels**, it fell back to **LLM + prompt engineering**, called per taxonomy level.
- Extended the classifier into a **Portfolio Benchmarking** feature for customers with many patents but minimal labels: **embedding-based proximity search** to taxonomy categories selected **class-balanced candidates** for LLM labeling (avoiding random sampling so imbalanced classes trained well); the LLM-labeled subset bootstrapped supervised training, which then ran inference on the rest — containing LLM cost and latency. Both classifier systems are protected by **published USPTO applications** (US20260212116A1; US20250272608A1).
- Ran controlled comparisons of SFT / LoRA / QLoRA / continued pretraining across Llama, Gemma, and Qwen variants; deployed best configurations via AWS DLC and MCP servers, with integrations to Amazon Bedrock, Google Gemini, and Azure AI.
- Trained and fine-tuned diffusion models for image generation and editing: dataset curation, conditioning design, training stabilization, and inference optimization to meet latency, cost, and visual-quality targets.
- Built interpretability tooling for transformer models (attention attribution, contrastive explanations) to meet transparency / traceability requirements for enterprise customers.
- Owned the **experimentation, training, evaluation, and deployment** lifecycle of production ML services; collaborated closely with **DevOps** on **CI/CD integration** and endpoint productionization; drove code-review, testing, and clean-code standards across the team.

Double Avatars

Spain

AI Engineer

Feb. 2023 – Aug. 2023

- Developed virtual try-on and 3D avatarization models for e-commerce, combining generative image models with body-shape estimation to improve user engagement and personalization.
- Streamlined end-to-end training and inference pipelines for computer vision models, increasing reliability and runtime efficiency.
- Collaborated with UX/UI and backend teams to integrate ML services into production systems and iterate on user experience through experimentation.

Freelance

Spain

Freelance Data Scientist

Sep. 2022 – Feb. 2023

- Enhanced semantic search engines, improving **MRR@15 by 8%** via iterative model and ranking-feature improvements.
- Built multimodal price-estimation and recommender systems, **reducing MAE by 12%** by combining vision and text signals with robust evaluation.
- Mentored 20+ students in applied ML through structured projects, code reviews, and deployment best practices.

Semrush

Spain

Data Scientist

May 2022 – Aug. 2022

- Developed active-learning NLP pipelines on large-scale web text to accelerate high-quality data labeling.
- Built and deployed NLP models to extract business insights from unstructured text.

CafeBazaar

Tehran, Iran

Data Scientist

Nov. 2020 – May 2022

- Implemented and productionized semantic app-search models, **increasing Recall@20 by 9%** for a large-scale Persian marketplace.
- Built automated POI extraction pipelines with YOLO, OCR, and SATRN, **raising F1 from 53% to 65%** while halving inference time.
- Optimized multi-GPU training for segmentation, **+11% mIoU** on vehicle-pricing use cases and reducing time-to-train.
- Developed activity-recognition systems that significantly reduced error rates in behavioral-understanding features.
Twice promoted within one year for outstanding performance.

Baharan

Iran

Data Scientist

Feb. 2020 – Oct. 2020

- Developed facial analysis / recognition models for surveillance systems.
- Shipped new facial-attribute detection features that expanded product capabilities and real-time use cases.

- **Yousefiramandi, A.** & Cooney, C. (2026). *When Does Synthetic Patent Data Help? Volume-Fidelity Trade-offs in Low-Resource Multi-Label Classification*. arXiv:2605.24297. **[first author]** — Submitted to ACL Rolling Review; target venue: EMNLP 2026.
- **Yousefiramandi, A.** & Cooney, C. (2026). *Benchmarking Patent Embeddings: A Multi-Task Evaluation of 22 Models Across Retrieval, Classification, and Clustering*. arXiv:2605.24296. **[first author]** — Submitted to ACL Rolling Review; target venue: EMNLP 2026.
- **Yousefiramandi, A.** & Cooney, C. (2025). *Patent Language Model Pretraining with ModernBERT*. arXiv:2509.14926. **[first author, 4 citations]**
- **Yousefiramandi, A.** & Cooney, C. (2025). *Fine-Tuning Causal LLMs for Text Classification: Embedding-Based vs. Instruction-Based Approaches*. arXiv:2512.12677. **[first author]**
- **Yousefiramandi, A.** (2026). *Cumulative-Goodness Free-Riding in Forward-Forward Networks: Real, Repairable, but Not Accuracy-Dominant*. arXiv:2605.06240. **[sole author]** — Submitted to NeurIPS 2026.
- **Yousefiramandi, A.** (2026). *Learning Selective Merge Policies for Deadline-Constrained Coded Caching via Deep Reinforcement Learning*. arXiv:2605.15236. **[sole author]** — Submitted to IEEE ICNP 2026; **40.9%** reduction in broadcast-packet expiration ratio.
- **Patent (pending — in verification):** *Prior Art Search Using Artificial Intelligence Agents*. Camelot UK Bidco Ltd. / Clarivate; Docket No. I03.00450000. **[sole inventor]**
- **Patent (published):** *Training an Artificial Intelligence Classifier Using LLM-Labeled Training Data*. U.S. Application No. 19/029,628 (Pub. US20260212116A1, 23 Jul. 2026); PCT counterpart WO/2026/154406, published 23 Jul. 2026. Docket No. I03.00430000 (Camelot UK Bidco Ltd.). **[first inventor]**
- **Patent (published):** *Hybrid Artificial Intelligence Classifier*. U.S. Application No. 18/590,188 (Pub. US20250272608A1); PCT counterpart WO/2025/181626, published 4 Sep. 2025. *Co-inventor*.

SELECTED OPEN-SOURCE PROJECTS

Text-JEPA Experiments & AF-JEPA | *PyTorch, MiniLM, Qwen 2.5-7B* Open-source

- Two empirical studies on JEPA target design for text representations: semantic-card ablation across 4 seeds (vs. raw embeddings / summaries / shuffled controls) on WikiText-2; **AF-JEPA** swaps the EMA target encoder for a frozen pretrained LLM (Qwen 2.5-7B), trained with smooth-L1 alignment on FineWeb-Edu.
- Block-masking (I-JEPA style) + InfoNCE / cosine alignment; raw-embedding target reaches **0.242 ± 0.016** top-1 retrieval vs. semantic cards **0.183 ± 0.008**. GitHub.

SSM Benchmark: Mamba-2 vs. Transformer vs. Jamba | *PyTorch, CUDA, FineWeb-Edu* Open-source

- Apples-to-apples training and inference benchmark of selective state-space models (**Mamba-2**), a Llama-style **Transformer** (RoPE + SwiGLU), and hybrid **Jamba** (Mamba + attention + MoE) on a ~1.5B-token FineWeb-Edu slice with GPT-2 BPE.
- Evaluated on MQAR, induction copy with extrapolation, and selective copy; measures validation perplexity, throughput, peak memory, and decode latency vs. sequence length. Confirms Mamba’s flat decode latency / KV-cache memory vs. Transformer growth, with Jamba in between. Resumable training (atomic checkpoint of model / optimizer / scheduler / RNG / data cursor) survives Colab’s 12-hour session cap. GitHub.

Multilingual LLM Adaptation via MAD-X Adapter Stacking (Swahili Biomedical) | *HF Adapters, AfroXLMR* Open-source

- Composable adapter stack on a frozen **AfroXLMR-large** backbone: Swahili language adapter + English biomedical domain adapter (MLM-trained) + supervised task adapters, stacked at inference (MAD-X / Pfeiffer).
- Targets medical MCQA (MMLU-ProX-Swahili), Swahili NER (MasakhaNER, BC5CDR-Disease projected), and topic classification; training data from NLLB-200 and OpenWHO. GitHub.

Instruction-Tuned LLM for Text Classification (LoRA/QLoRA) | *TRL, PEFT* Open-source

- Instruction-tunes a decoder-only LLM to output exact label strings; LoRA adapters + 4-bit quantization (QLoRA-style); demo reports **~94% accuracy**. GitHub | Model.

Multi-Modal Hateful Content Classification (CLIP/SigLIP Fusion + MTL) | *CLIP, SigLIP* Open-source

- Late-fusion gated attention + per-label multi-task head on MMHS150K; test **macro-F1 0.569, ROC-AUC (macro) 0.783**. GitHub | Model.

Text+Image VLM Fine-Tuning via JSON Generation | *Qwen2-VL, PaliGemma2* Open-source

- Generative classification on MMHS150K: prompt the VLM to return JSON, parse extracted labels, compute multi-label metrics; Qwen2-VL+LoRA **micro-F1 0.611** (test). GitHub | Adapter.

Whisper-based Multi-Task ASR + Voice Intensity | *PyTorch, Transformers, SageMaker* Open-source

- Joint speech recognition + loudness regression on Whisper encoder; multi-task loss $\mathcal{L} = \mathcal{L}_{\text{ASR_CE}} + \lambda \cdot \text{MSE}$.
Reported: **WER 4.6976%**, intensity **RMSE 0.7334**. GitHub | Model.

Wav2Vec2 Multi-Task: CTC ASR + Intensity | *PyTorch, HF, SageMaker* Open-source
 • Wav2Vec2-base-960h CTC + intensity head with attention-masked mean pooling; validation **WER $\approx 12.9\%$** ; config-first repo. GitHub.

Image-Text Retrieval (CLIP/SigLIP) | *FAISS-ready, custom Trainer* Open-source
 • Contrastive training/eval on Flickr8k/30k: Flickr8k **R@1 90.7%** (Img→Text), **77.06%** (Text→Img); Flickr30k **R@1 92.3%** / **79.0%**. GitHub.

Multi-Label Emotion Classification (GoEmotions) | *Transformers* Open-source
 • 27-emotion multi-label detection with RoBERTa-base; validation threshold tuning; test **micro-F1 0.528 @ thr 0.84, ROC-AUC (micro) 0.952**. GitHub.

Emotion-Aware TTS Style Transfer | *SpeechT5, WavLM, ECAPA, HiFi-GAN* Open-source
 • StyleAdaptor + StyleSpeakerFusion: fuse WavLM emotion/prosody with ECAPA speaker identity for expressive TTS; Gradio UI. GitHub.

TECHNICAL SKILLS

Research / Modeling: Transformer architectures (MHA, GQA, MoE), attention variants (Flash, paged), positional encodings (RoPE, ALiBi), state-space models (Mamba, Mamba-2, Jamba), RLHF / DPO / GRPO, knowledge distillation, continued pretraining, tokenizer extension, controlled ablation design

Frameworks: PyTorch, Hugging Face (Transformers, TRL, PEFT, Diffusers), MosaicML Composer, MLflow, Scikit-learn, LightGBM

LLMs & Agents: UnSloth (Llama / Gemma), LoRA / QLoRA, DSPy, DeepAgents, LangGraph, LlamaIndex, MCP, Strands Agents, AgentCore, Google ADK, DeepEval

Cloud: AWS (SageMaker, Glue, Lambda, Fargate, Bedrock, Step Functions), Azure AI, Google Gemini integrations

Languages & Tools: Python, C++, Bash, SQL, Git, Docker, ONNX, Jenkins, L^AT_EX

Quantitative Research: Mathematical optimization, mathematical modeling, statistical learning, information theory

Spoken Languages: English (fluent), Spanish (working), Persian (native)

EDUCATION

Sharif University of Technology	Tehran, Iran
<i>M.Sc. in Electrical Engineering (AI)</i>	2018 – 2021
Isfahan University of Technology	Isfahan, Iran
<i>B.Sc. in Electrical Engineering</i>	2014 – 2018

HONORS AND AWARDS

- Promoted at Clarivate within less than a year for exceptional performance.
- Twice promoted at CafeBazaar within one year.
- Ranked 16th nationwide in Iran’s master’s-program entrance examination.